

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Saturday, 12 September 2026, 9:54 PM**State** Finished**Completed on** Saturday, 12 September 2026, 9:56 PM**Time taken** 2 mins 26 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Chessboard:**

Ram is given with an $n \times n$ chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position ($n-1$, $n-1$) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:**Input**

```
3
1 2 4
2 3 4
8 7 1
```

Output:

```
19
```

Explanation:

Totally there will be 6 paths among that the optimal is
Optimal path value: $1+2+8+7+1=19$

Input Format

First Line contains the integer n
The next n lines contain the $n \times n$ chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6     scanf("%d", &n);
7
8     long long a[n][n];
9     long long dp[n][n];
10
11     for (int i = 0; i < n; i++)
12     {
13         for (int j = 0; j < n; j++)
14         {
15             scanf("%lld", &a[i][j]);
16         }
17     }
18
19     dp[0][0] = a[0][0];
20
21     // First row
22     for (int j = 1; j < n; j++)
23         dp[0][j] = dp[0][j - 1] + a[0][j];
24
25     // First column
26     for (int i = 1; i < n; i++)
27         dp[i][0] = dp[i - 1][0] + a[i][0];
28
29     // Remaining cells
30     for (int i = 1; i < n; i++)
31     {
32         for (int j = 1; j < n; j++)
33         {
34             if (dp[i - 1][j] > dp[i][j - 1])
35                 dp[i][j] = a[i][j] + dp[i - 1][j];
36             else
```

```
37         dp[i][j] = a[i][j] + dp[i][j - 1];
38     }
39 }
40
41 printf("%lld", dp[n - 1][n - 1]);
42
43 return 0;
44 }
```

	Input	Expected	Got	
✓	3 1 2 4 2 3 4 8 7 1	19	19	✓
✓	3 1 3 1 1 5 1 4 2 1	12	12	✓
✓	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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